



GROUP TECHNICAL SOLUTIONS (GTS)

TECHNICAL REPORT FOR PETRONAS CHEMICAL LOW DENSITY POLYETHYLENE SDN BHD (PCLDPE)

PCLDPE Base Layer Controller Assessment

February 2020

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	Name:	Signature:	Role:	Date:
Prepared by:	Muhammad Aminuddin Ishak		Executive Process Control	
Reviewed by:	Andriny Rusman		Principal Process Control	
Approved by:	Fauziah Ali		Custodian Process Control	

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1 EXECUTIVE SUMMARY

PETRONAS Chemicals Low Density Poly Ethylene Sdn Bhd (PCLDPE) have embarked an Advanced Process Control revamp project. Prior to APC Implementation, a Base Layer Controller and PIMS Assessment and Rectification was conducted. This assessment was carried out on 42 controllers which are APC (34 controllers) and GTM (8 controllers) related controllers. The data for controllers assessment was collected from 1st December until 31st December 2019, a month after PCLDPE start up in October 2019. In addition to the base layer controllers, the quality of data retrieved via PCLDPE plant information management system (PIMS) which is PI was assessed and data with issues are identified

For APC related controllers, out of 34 controllers, 9 of them are APC MV controllers. The remaining controllers mainly comprises of controllers located within the APC optimization process unit. The majority of the assessed controllers are from Unit 12 (Polymerization) with 30 controllers while 1 and 3 for Unit 11 and Unit 13. The 9 APC MVs are located in Unit 11 (1) and Unit 12. (8). 94% (32) of controllers assessed are operated in AUT mode, where only 6% are operated in MAN. According to PTS 16.43.01, the criteria for Good Rating is 80%, and PCLDPE Base Layer Controller can be rated as GOOD rating. The controllers that are in MAN mode are 12TIC1113 and 11FIC0501. While for controller compliances, 94% of 34 controllers are in complying mode. As APC MVs is the main handle to the process, it is important for the controllers to be operated in design mode. From the analysis, 100% of 9 MVs are operated in AUTO mode.

For GTM related controllers, A total of 8 GTM related controllers had been assessed which come from 3 different units which are Unit 11 –Compression (1) , Unit 13 – Palletising (6) and Unit 27 – Hot oil (1). However, due to unavailability of PI tags for several controllers, the analysis cannot be conducted for 2 of the controllers which are 13FIC0211 and 13FIC0212. 6 controllers assessed are all in compliance as all controllers can be operated in the design mode. Out of 6 controllers, 4 of them are good in controller performance. For another 2 controllers which are 13TIC0501 and 13FIC0606.

In conclusion, as APC related base layer controller's mode (94%) and compliance status (94%) achieved good rating as per PTS standards, generally, PC LDPE Base Layer Controllers are ready for APC Implementation.

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2 ABBREVIATION

AUT	Controller is operated is AUTO mode
MAN	Controller is operated is MANUAL mode
BLC	Base Layer Controller
APC	Advanced Process Control
MV	Manipulated Variable
PV	Process Variable
SV	Set point Variable
GTM	Grade Transition Manager

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3 INTRODUCTION

PETRONAS Chemicals Low Density Poly Ethylene Sdn Bhd (PCLDPE) have embarked an Advanced Process Control revamp project which consists of 3 components which are

- a. Polymer Inferential Properties
- b. Apollo Advanced Process Control
- c. Process Recipe and Transition Manager

Prior to APC Implementation, a Base Layer Controller and PIMS Assessment and Rectification was conducted. This assessment was carried out on 42 controllers which are APC and GTM related controllers. The data for controllers assessment was collected from 1st December until 31st December 2019, a month after PCLDPE start up in October 2019.

In addition to the base layer controllers, the quality of data retrieved via PCLDPE plant information management system (PIMS) which is PI was assessed and data with issues are identified and rectified.

4 OBJECTIVES

The objectives of BLCAR are as below:

1. To assess the status of the controllers and instrumentation in PCLDPE prior to APC Implementation
2. To evaluate and assess the performance of the controller
3. To identify and rectify issue found during the assessment

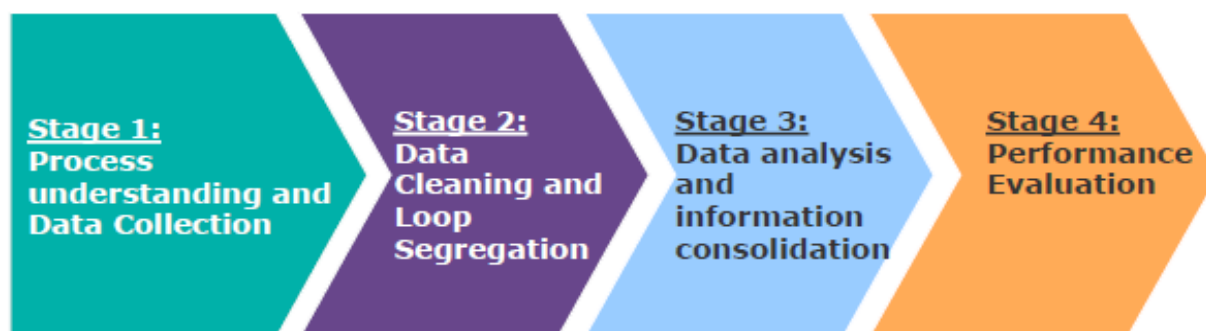
While the objectives on PIMS assessment are:

1. To ensure all essential tags used by base layer and APC for performance analysis, benefit analysis, monitoring and modelling are of good quality.
2. To ensure all essential tags used by base layer and APC for performance analysis, benefit analysis, monitoring and modelling are historized.

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5 METHODOLOGY

Based on PTS 160431, the methodology on BLCA are as below:



Stage 1: Process Understanding and Data Gathering

1. PFD & Process Description
2. P&ID
3. DCS Graphic
4. Controller Narratives
5. PID Tuning Parameters
6. Data (1 min interval) from PIMS
 - a. For this assessment, a period of 1 month is taken from 1st Dec 2019 at 00:00 hours to 31st December 2019 at 00:00 hours

Stage 2: Data Cleaning and Loop Segregation

1. Controller mode (Manual / Auto)
2. Controller Status (Compliance / Non Compliance)

Stage 3: Data Analysis

1. Interview with operators, technologist, and system engineer to understand the data collected.
2. Trend observations
3. Performance Analysis using MS Excel and Shell MDPro
4. Check basic controller configuration on DCS
5. Check PIMS data quality

Stage 4: Performance Evaluation

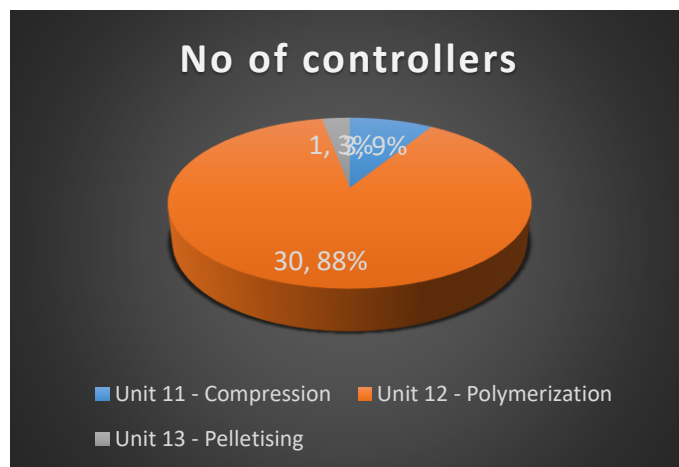
1. Determine controller performance
2. Provide recommendations and generate rectification list

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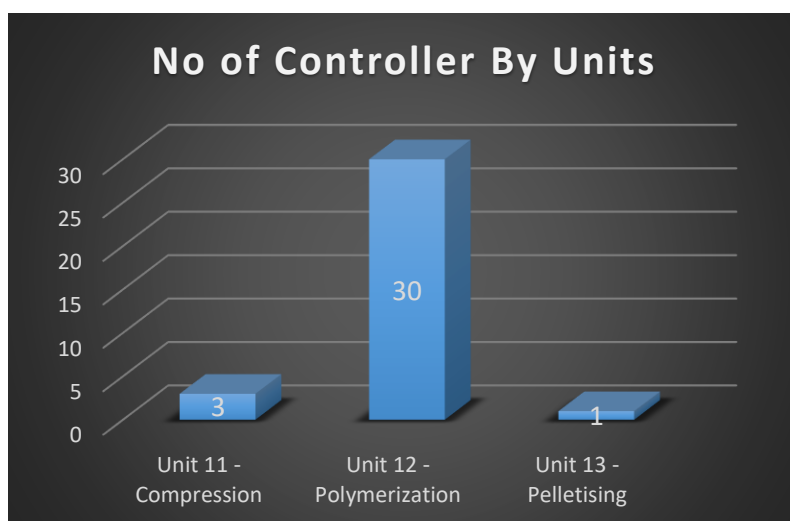
6 BASE LAYER CONTROLLERS ASSESSMENT FINDINGS AND RESULT

6.1 APC Related Controllers

i. Number of controllers Assessed



A total of 34 controllers were assessed. Where 9 of them are APC MV controllers. The remaining controllers mainly comprises of controllers located within the APC optimization process unit.

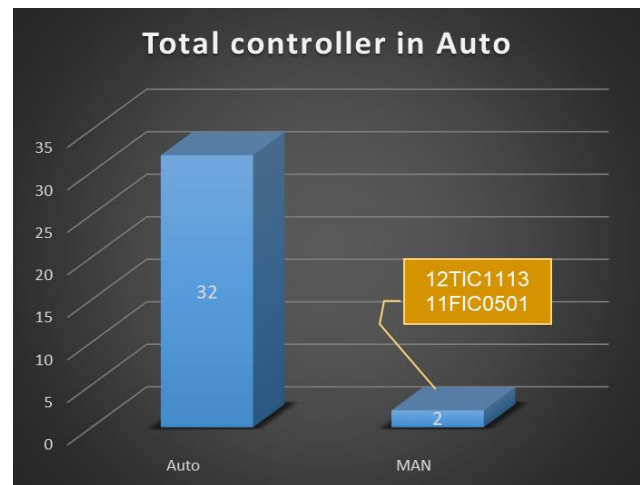
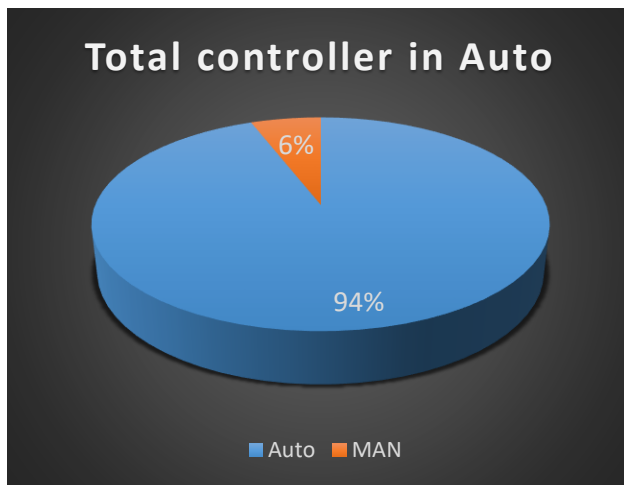


The majority of the assessed controllers are from Unit 12 (Polymerization) with 30 controllers while 1 and 3 for Unit 11 and Unit 13. The 9 APC MVs are located in Unit 11 (1) and Unit 12.(8) . Other controllers are related to APC in terms of optimization.

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ii. Controllers' Mode Assessment

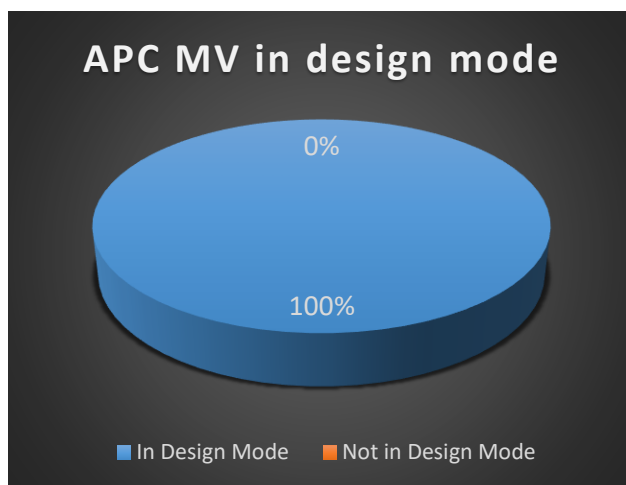
The controllers' mode here refers to the operating mode of the controller where it can be Manual (MAN), Automatic (AUT), Cascade (CAS) or RCAS. For simplicity, CAS/RCAS will be categorized as AUT in this report



Referring to above figures, 94% (32) of controllers assessed are operated in AUT mode, where only 6% are operated in MAN. According to PTS 16.43.01, the criteria for Good Rating is 80%, and PCLDPE Base Layer Controller can be rated as GOOD rating. The controllers that are in MAN mode are 12TIC1113 and 11FIC0501.

It was confirmed that 11FIC0501 is intentionally operated in MAN most of the time as it is for Propane flow. It will just be used whenever plant got issue with insufficient propylene. While for 12TIC1113, it has been operated in MAN due to process non linearity. 12TIC1113 is the outlet temperature of recycle gas into compressor. The process behaviour is highly depended on cooler dewaxing process. Dewaxing process is a daily cleaning routine for cooler conducted manually by panel operator. When the wax built up, it reduces the heat exchange efficiency and hence will impact the controller performance. As this procedure is conducted manually, it is exposed to human potential error which cause higher non linearity issue.

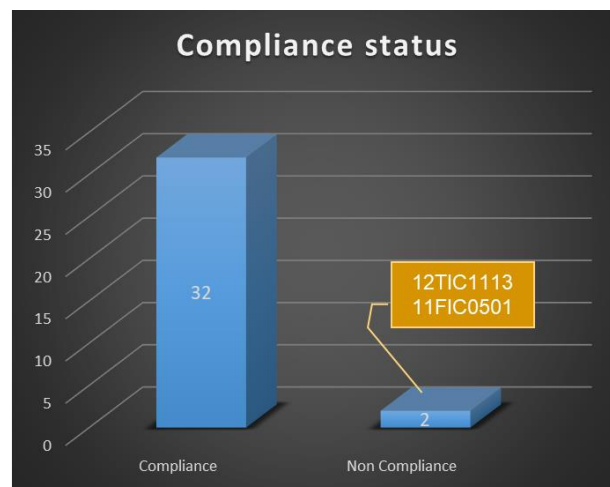
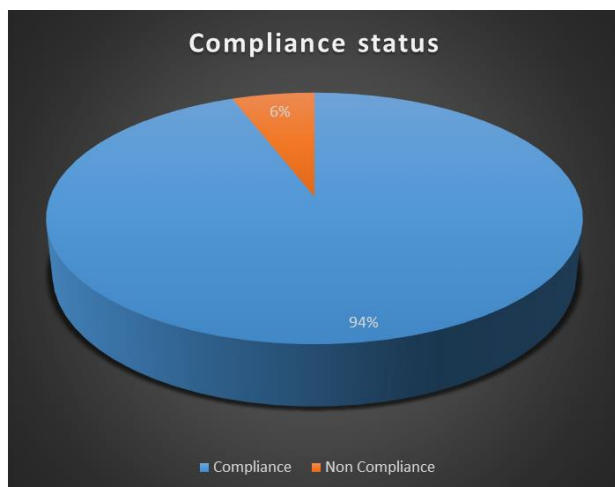
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Besides that, 100% of APC MV are already operated in design mode. This is good to ensure the APC can work as the MV are operated in AUT.

iii. Compliance Status Assessment

The controller compliance on its design mode is also assessed to identify if any of the controllers are currently not operated as per design. Normally, all controllers are design to be operated in AUT/CAS, however under some circumstances, it can be operated otherwise (eg in MAN). This finding is relevant to check if the controllers assessed are operating as per design intend.

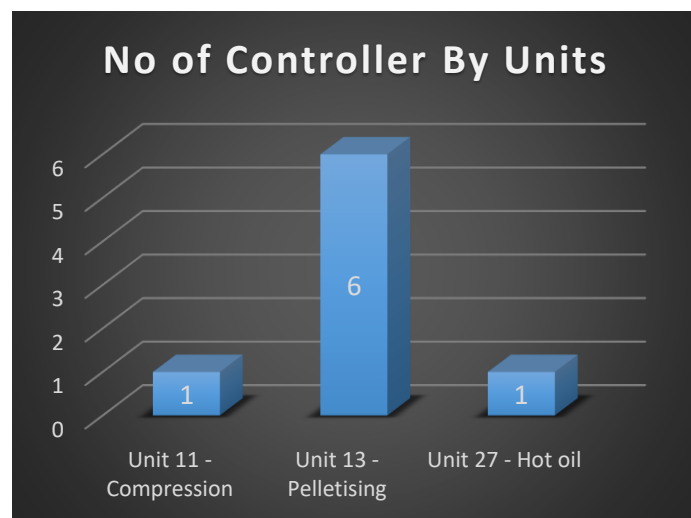
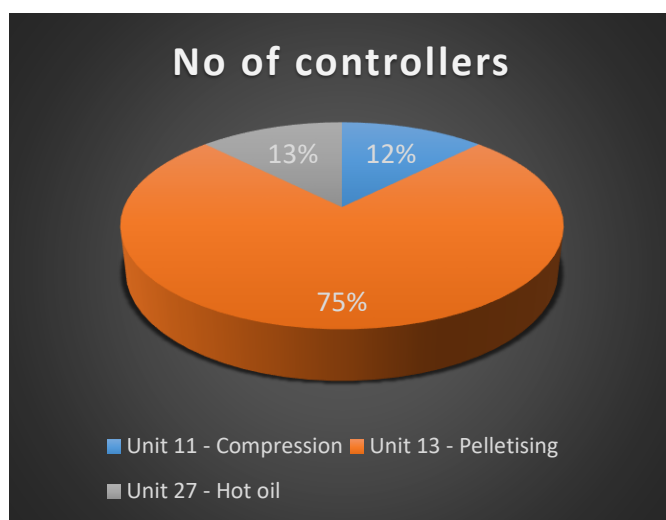


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Referring to the figure above, 94% of 34 controllers are in complying mode, According to PTS 16.43.01, this is rated as GOOD rating.

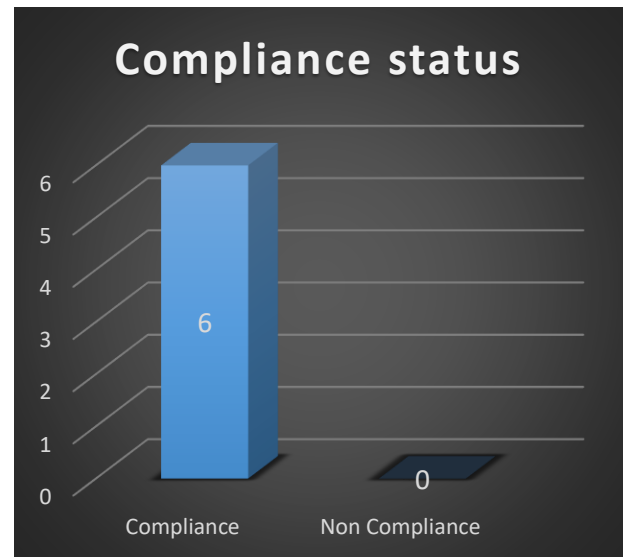
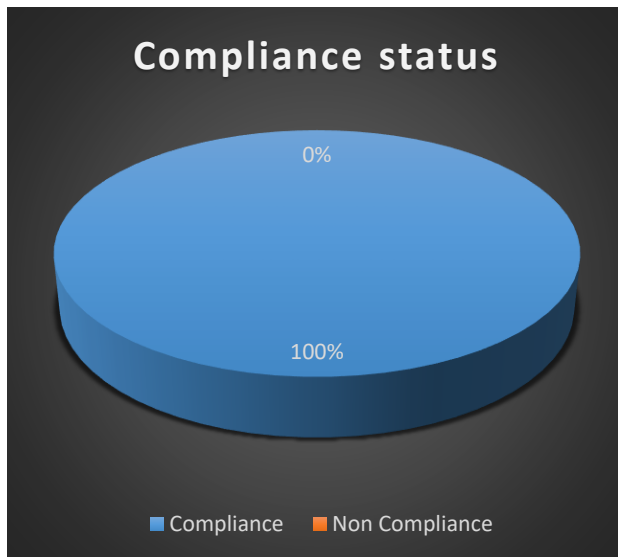
6.2 GTM Related Controllers

A total of 8 GTM related controllers had been assessed which come from 3 different units which are Unit 11 –Compression (1) , Unit 13 – Palletising (6) and Unit 27 – Hot oil (1)

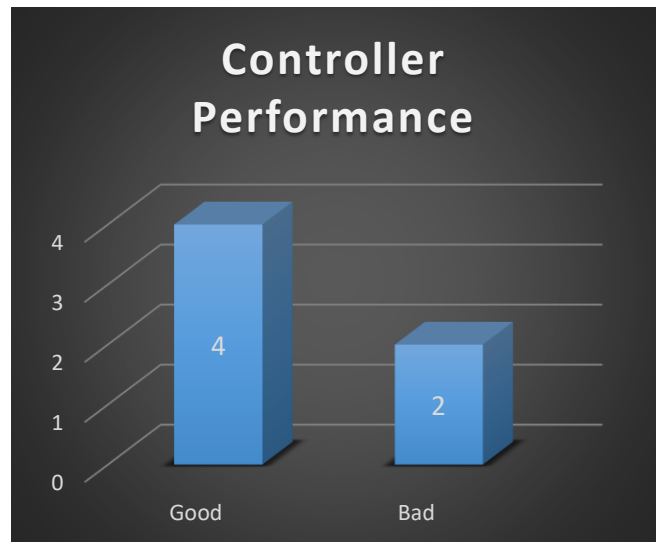
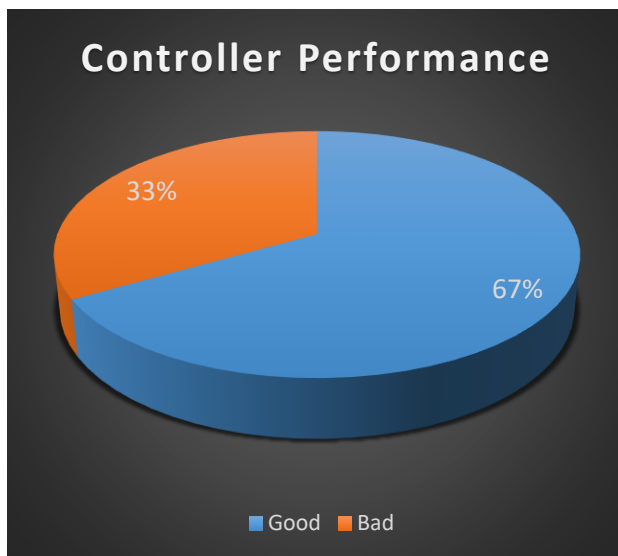


However, due to unavailability of PI tags for several controllers, the analysis cannot be concluded for 2 of the controllers which are 13FIC0211 and 13FIC0212.

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6 controllers assessed are all in compliance as all controllers can be operated in the design mode.

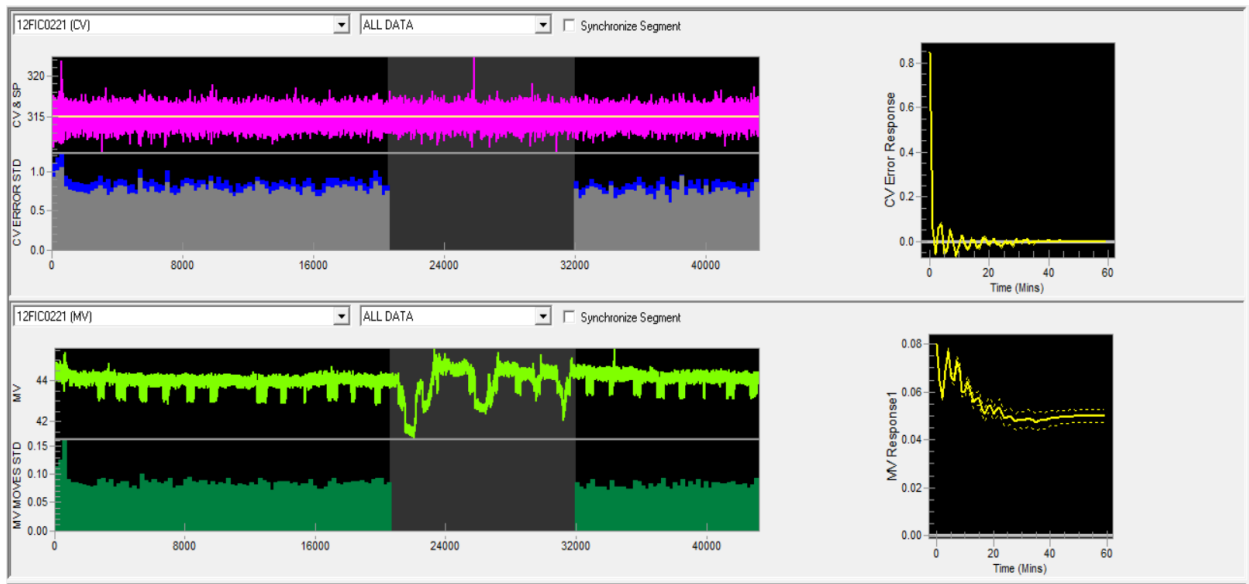


Out of 6 controllers, 4 of them are good in controller performance. For another 2 controllers which are 13TIC0501 and 13FIC0606, it was observed that its MV sometimes hit saturation either at 0% or 100%. It means the controllers lose its control when hitting saturation limit.

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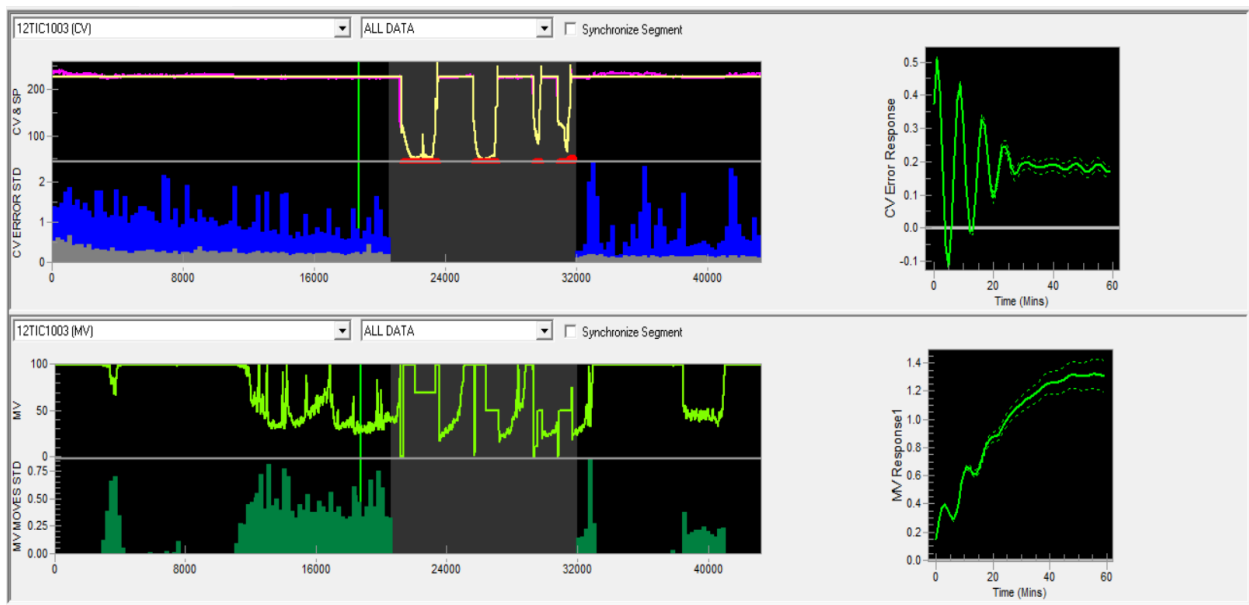
6.3 Highlight of Controller Performance

Controller with good performance – 12FIC0221



Good control was indicated by the size of blue area. The smaller the blue area, the better the controller.

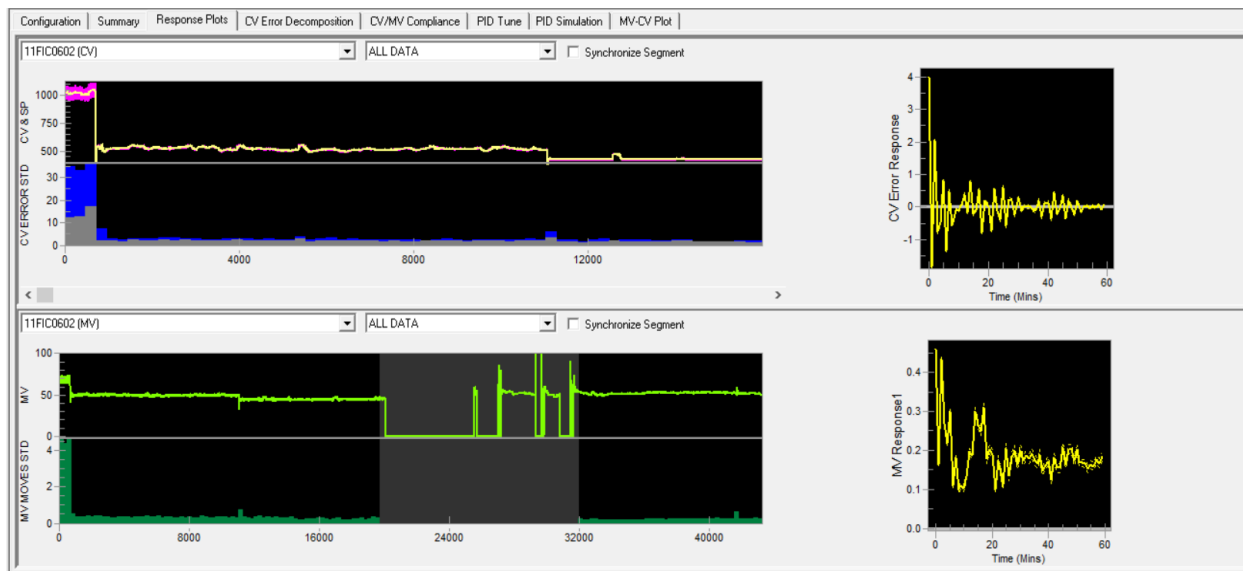
Controller with Poor Performance – 12TIC1003



This controller is saturated at 100% most of the time. Limited range of controllability. PV is floating.

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Controller with Potential Issue– 11FIC0602



CTA flow shows increase in standard deviation as flow increase. At higher flow (1020 kg/h), the standard deviation increase by 8 times. The standard deviation increment is from 2kg/h to 16kg/h

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7 RECOMMENDATION

- 1) To create SV and MV tags in PI server for controllers 13FIC0211 and 13FIC0212.
- 2) PC LDPE Instrument Engineer may look into reviewing transmitter design and any other possible cause for 11FIC0602
- 3) Gain scheduler may be can be configured for 11FIC0602 if the gain and dynamic at higher flow has big difference compared to another region.
- 4) PC LDPE is encouraged to introduce a consistent technique on cooler dewaxing to eliminate the possibility of process non linearity occurrence. This maybe can be done by process sequencer (i.e Exapilot), and thus enable the automation of 13TIC1113.
- 5) To configure filtered Peak temperature at DCS instead of in APC. This should help to reduce the amount of token used.

8 CONCLUSION

- 1) APC related base layer controllers mode (94%) and compliance status (94%) achieved good rating as per PTS standards
- 2) CTA flow 11FIC0602 measurement shows high variability as flow increase. This need major attention for rectification as MFI variation is very sensitive to CTA flow variation (0.2 MFI = 10 kg/h CTA flow)
- 3) Generally, PC LDPE Base Layer Controllers are ready for APC Implementation.